

☐ Revisit Later

Select an option

☐ Partial Dependency☐ Transitive Dependency☐ Fully Functional Dependency☐ Multivalued Dependency

Question 140

Consider the given relation: $GH \rightarrow I$, $I \rightarrow G$, $H \rightarrow G$

What kind of dependency is present in this relation? Select the correct answer from the given choices.

Patel Kalpit Dineshk-124592599

Question 1

Assume that a system administrator wants to set up a file server for his/her organization. He/she wants to ensure that each department has access only to its own files and folders on the server. Which feature of the operating system can help the admin achieve this goal?

Revisit Later

Select an option

- ☐ Access Control Lists (ACLs)
- ☐ User accounts
- ☐ File ownership
- ☐ File permissions inheritance

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5. OOPS

Question 1

Revisit Later

Select an option

As a software architect, assume that you are designing a new system that uses object-oriented programming principles. You want to ensure the system is flexible, scalable, and easy to maintain. In object-oriented programming, which is not a fundamental principle of the SOLID design principles? Select the correct answer from the given choices.

☐ Single Responsibility Principle☐ Open-Closed Principle☐ Liskov Substitution Principle☐ Program Execution Principle

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1 2 3 4 5

Attempted: 0/5

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Select an option

☐ Multivalued attributes☐ No atomicity☐ Partial dependency☐ No partial dependency

Question 2

Consider the given relations with functional dependencies:

 $PQ \rightarrow R$ $R \rightarrow S$ $R \rightarrow T$ $T \rightarrow U$ $U \rightarrow P$

Which aspect is present in the given relations? Select the correct answer from the given choices.

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Question 2

Assume that you are given the task to modify your company's operating system so that the context switching of the CPU becomes faster. Which feature will you modify so that you can accomplish the given task? Select the correct answer from the given choices.

Select an option

☐ CPU scheduling

☐ Memory disk

☐ Processes

☐ Thread

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3. Database Concept

1

2

3

4

5

Previous

Next

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Question 2

Consider the given relations with functional dependencies:

 $P \rightarrow R$ $R \rightarrow S$ $R \rightarrow T$ $T \rightarrow U$ $U \rightarrow P$

Which aspect is present in the given relations? Select the correct answer from the given choices.

Select an option

☐ Multivalued attributes☐ No atomicity☐ Partial dependency☐ No partial dependency

5. OOPS

Question 2

In a vehicle management system, assume that you have the base class "Vehicle" with a property "price". When the subclass "Car" inherits from the "Vehicle" class, what happens to the "price" property in the "Car" class? Select the correct option from the given choices.

Revisit Later

Attempted: 0/7

Select an option

Clear Response

- ☐ The subclass gains access to the public and protected members of the superclass
- ☐ The subclass only gains access to the private members of the superclass
- ☐ The subclass does not gain access to any members of the superclass
- ☐ The subclass gains access to only the static members of the superclass

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6. Operating System

1 2 3 4 5 6

Previous

Question 3

Assume that you are working on a project in which you have to support an array of behavior, including timesharing multiple processes so that various tasks are available. Which type of management will you choose to accomplish this task? Select the correct answer from the given choices.

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Select an option

☐ Application management☐ Memory management☐ Time stamping management☐ Disk management

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Question 3

Consider that there are 5 attributes present about R. They are PQRS. Fields of R contain only atomic values.

$R = \{PQ \rightarrow R, R \rightarrow S, P \rightarrow T, Q \rightarrow R, R \rightarrow P, PR \rightarrow Q\}$ is a set of functional dependencies.

How many candidate keys does the relation R have? Select the correct answer from the given choices.

Revisit Later

Attempted: 0/5

Select an option

☐ 3

☐ 4

☐ 2

☐ 7

Question

Assume that you want to store an image as a byte array with the help of the given code:

Revisit Later

Select an option

@Large

@Lob

○ @Object

○ @LargeObj

```

1 @Entity
2 class Image{
3
4     private String id;
5
6     private String fileName;
7
8     private String fileType;
9
10    private byte[] imgData;
11 }

```

Which annotation can help the byte array in storing large objects? Select the correct answer from the given choices.

S. OOPS

Question 3

Revisit Later

Attempted: 0/7

Select an option

What will be the output of the given Java Code?

```
1 public class Max{
2     static int addMethod (int a, int b) {
3         return a+b;
4     }
5
6     static double addMethod(double a, double b) {
7         return a + b;
8     }
9
10    public static void main(String[] args) {
11        int x = addMethod(8, 5);
12        double y = addMethod(4.3, 6.26);
13        System.out.println(x+y);
14    }
15 }
```

☐ Compile-time Error☐ Run-time Error☐ 23.0☐ 23.56

Select the correct answer from the given choices.

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5. OOPS

< 1 2 3 4 5 6 7 >  

Attempted: 0/7

Question 4

☐ Revisit Later

Select an option

Suppose you are designing a simple banking application using Object-Oriented Programming (OOPS) concepts.

Then, identify which of the following concepts would be best suited to represent a customer's bank account?

☐ Class☐ Object☐ Inheritance☐ Polymorphism

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Question
Assume that four processes, A, B, C, and D, share a database resource. Process A holds a read lock on a table and waits for a write lock. Process B holds a write lock on the table and waits for a read lock. Process C holds a read lock on another table and waits for the write lock held by Process B, while Process D holds a write lock on yet another table and waits for the read lock held by Process A. The given scenario denotes an example of which situation? Select the correct answer from the given choices.

Revisit Later

Select an option

- ☐ Time slice exhaustion
- ☐ Livelock
- ☐ Resource depletion
- ☐ Circular wait

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Question 4

Consider that you are the database administrator of a large online retail store. You need to ensure that the data entered into the database is consistent and accurate. Which concept of DBMS will you use to fulfill this requirement? Select the appropriate answer from the given choices.

☐ Revisit Later

Select an option

☐ Redundancy

☐ Data Integrity

☐ Atomicity

☐ Data Independence

Question

Assume that a database contains a table with the given attributes: "Student ID", "Student Name", "Student Address", and "Student Phone Number". Consider the table is prone to data redundancy and data anomalies. To resolve these issues, which normalization form should the table be in? Select the appropriate answer from the given choices.

Select an option

- ☐ Fourth Normal Form (4NF)
- ☐ Third Normal Form (3NF)
- ☐ First Normal Form (1NF)
- ☐ Second Normal Form (2NF)

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Question

Assume that you are working with a group of processors who do not share a memory or other peripherals, and all the processors communicate through the network. Which feature is a characteristic of this type of processor architecture? Select the correct answer from the given choices.

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Select an option

☐ Shared memory☐ Symmetric multiprocessing☐ Distributed processing☐ Cache coherence

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Question 510

As a human resources manager at a large corporation, consider that you want to store information about the company's employees in a database management system. Which resource will be most useful to ensure that each employee has a unique identifier in the database? Select the correct answer from the given choices.

☐ Revisit Later

Select an option

- ☐ Composite Key
- ☐ Candidate Key
- ☐ Foreign Key
- ☐ Primary Key

Question 5

Consider that you are designing a new healthcare organization app to store patient data in a database management system. Which task will be most useful for the healthcare organization to ensure that the patient data remains confidential and secure? Select the correct answer from the given choices.

☐ Revisit Later

Select an option

☐ Data Retrieval

☐ Data Updation

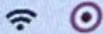
☐ User Administration

☐ Data Definition

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5. OOPS

Question 5

Assume that you are developing a software system for a car rental company. Which option best describes the use of abstraction in this scenario? Analyze the given choices and select the correct option.

Revisit Later

Select an option

Clear Response

☐ Abstraction is used to represent only the essential features of a car such as its make, model, and rental price while hiding unnecessary details such as its engine capacity and internal components.

☐ Abstraction is used to represent all features of a car, including make, model, rental price, engine capacity, and internal components.

☐ Abstraction is used to hide all the features of a car, including essential details such as its color and rental price.

☐ Abstraction is used to hide only essential details such as its

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Question 6

Consider that you are working on your computer system and suddenly, the performance of your computer decreases and it starts lagging. What action will you choose to increase the performance of your computer and enable the applications to run faster? Select the correct answer from the given choices.

Revisit Later

Select an option

- ☐ Threading
- ☐ Multithreading
- ☐ Scheduling
- ☐ Processes

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5 OOPS

< 1 2 3 4 5 6 7 >

Attempted: 6/7

Question 6

Revisit Later

Select an option

Clear Response

Assume that a software designer wants to create a class to represent an employee. The designer wants to ensure that the employee's personal information such as name, salary, and address can only be accessed and updated through methods provided by the class. Which OOP principle can the designer apply in this scenario? Select the correct option from the given choices.

☐ Polymorphism☐ Inheritance☐ Encapsulation☐ Abstractionn

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5 OOPS

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1

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9

Previous

Next Section

Question 7

Attempted: 0/7

Revisit Later

Select an option

Clear Response

Consider that ABC is an interface that declares two methods: myMethod() and getInfo(). xyz is an abstract class that implements the ABC interface and defines a method called getData(). Test is a concrete class that extends the xyz class and implements the myMethod() and getInfo() methods. When the main method is executed, a new Test object is created and assigned to a variable of type xyz. The getInfo() method is then called on this object, which outputs the string "xyz" to the console. The Test class also provides the implementation for the myMethod() and getInfo() methods inherited from the ABC interface. The myMethod() method outputs the string "How are you" to the console and the getInfo() method is implemented to output the string "xyz" to the console. Overall, this code demonstrates the use of inheritance and interfaces in Java to provide a common interface for related classes and objects, while allowing for flexibility in implementation details.

Which is the best code to print "XYZ" using polymorphism? Analyze the given choices and select the correct option.

```
1 interface ABC
2 {void myMethod();void getInfo();}
3 }abstract class abc implements xyz
4 {void getData(){System.out.println("Hello
World");}}
5 }public class Test extends xyz
6 {public void myMethod(){
7 System.out.println("How are you");}
8 public void getInfo(){
9 public static void main(String[] args)
10 {xyz obj = new Test();obj.getInfo();}}
```

```
1 interface ABC
2 {void myMethod();void getInfo();}
3 }abstract class abc implements
tData(){System.out.pri
}
```

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1. Coding Skills



Revisit Later

Question 1

How to Attempt?

Pass the Ball

A group of N players are playing a game. In this game, a player needs to pass the ball to another player. A player P holds the ball at the beginning of the game. A maximum of X moves are allowed while passing the ball such that it ends up with the same player who started the game. Given below is the condition that must be followed by all the players while passing the ball:

- A player K_1 can pass the ball to another player K_2 if K_1 divides K_2 or if K_2 divides K_1 .

Your task is to find and return integer value representing the number of possible ways to complete the game.

Note: A game is considered as "complete" if the ball ends up with the player who started it.

Input Specification:

input1 : An integer value N , representing the number of players.

input2 : An integer value P , representing the player who starts the game.

input3 : An integer value X , representing the maximum number of moves allowed to pass the ball.

Output Specification:

Return an integer value representing the number of possible ways to complete the game.

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Note: A game is considered as "complete" if the ball ends up with the player who started it.

Input Specification:

input1 : An integer value N, representing the number of players.

input2 : An integer value P, representing the player who starts the game.

input3 : An integer value X, representing the maximum number of moves allowed to pass the ball.

Output Specification:

Return an integer value representing the number of possible ways to complete the game.

Example 1:

input1: 3

input2: 2

input3: 2

Output: 1

Explanation:

There are 3 players in the game and, initially, the 2nd player has the ball. There is a maximum of 2 moves allowed, so the game can be completed in the following way:

- 2->1->2 Here, player 2 can pass the ball to player 1 as 1 divides 2. For the same reason, player 1 can pass the ball to player 2.

Since there is only 1 possible way to complete the game, 1 is returned as the output.

Example 2:

Output: 1

Explanation:

There are 3 players in the game and, initially, the 2nd player has the ball. There is a maximum of 2 moves allowed, so the game can be completed in the following way:

- 2->1->2 Here, player 2 can pass the ball to player 1 as 1 divides 2. For the same reason, player 1 can pass the ball to player 2.

Since there is only 1 possible way to complete the game, 1 is returned as the output.

Example 2:

input1: 3

input2: 2

input3: 4

Output: 3

Explanation:

There are 3 players in the game and, initially, the 2nd player has the ball.

There is a maximum of 4 moves allowed, so the game can be completed in the following ways:

- 2->1->2
- 2->1->3->1->2
- 2->1->2->1->2

Therefore, 3 is returned as the output.

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```
CPP17
Compiler: g++ 9.2.0

1  #include<stdio.h>
2  #include<string.h>
3  #include<string>
4
5  using namespace std;
6  // Read only region start
7
8  int passTheBall(int input1,int input2,int input3)
9  {
10     // Read only region end
11     // Write code here
12     throw "Function passTheBall(int input1,int input2,int input3) not implemented. ";
13 }
14
15
```

I


```
3  #include<bits/stdc++.h>
4  using namespace std;
5  int solve(int n, int p, int i, int a){
6      if(i == 0){
7          return (a==p);
8      }
9      int ans = 0;
10     for(int j = 1; j <= n; ++j){
11         if(j == a)
12             continue;
13         if(j % a == 0 || a % j == 0){
14             ans += solve(n, p, i-1, j);
15         }
16     }
17     return ans + (a == p);
18 }
19
20 int main(){
21     int n = 3;
22     int p = 2;
23     int x = 4;
24     cout << solve(n, p, x, p)-1;
25
26 }
27
```


Question 1

Report Error

How to Attempt?

State of the Art

Being a painter, Bruce thought of a new and futuristic idea for his artwork. He has X different paints on a palette and wants to create a painting with new paint combinations of the existing paints. He can utilize at most Y different paints for crafting a new paint combination.

Your task is to help Bruce find the number of different paint combinations he can craft from the existing paints on the palette and return an integer value representing the number of paints (new combinations and the existing paints).

Note:

- The volume of paint does not matter.
- Return answer modulo 7.

Input Specification:

input1 : An integer value X , representing the number of different paints on the palette.

input2 : An integer value Y , representing the maximum number of different paints he can use to create a new paint combination.

Output Specification:

Return an integer value representing the number of paints (new combinations and the existing paints).

Input Specification:

input1 : An integer value X, representing the number of different paints in the palette.

input2 : An integer value Y, representing the maximum number of different paints he can use to create a new paint combination.

Output Specification:

Return an integer value representing the number of paints (new combinations and the existing paints).

Example 1:

input1 : 5

input2 : 2

Output : 15

Explanation:

The existing paints and the number of paint combinations Bruce can make with 5 different paints using at most 2 different paints are:

- Existing Paints: (1), (2), (3), (4), (5)
- New Paint Combinations: (1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5).

As he can make 10 new paint combinations and the existing paints are 5, hence $10 + 5 = 15$ is returned as the output.

Example 2:

```
1 // Write your solution here
2 // Read only region starts
3 // Read only region ends
4 // Write your solution here
5 // Read only region starts
6 // Read only region ends
7 // Write your solution here
8 // Read only region starts
9 // Read only region ends
10 // Write your solution here
```

Output : 15

Explanation:

The existing paints and the number of paint combinations Bruce can make with 5 different paints using at most 2 different paints are:

- Existing Paints: (1), (2), (3), (4), (5)
- New Paint Combinations: (1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5).

As he can make 10 new paint combinations and the existing paints are 5, hence $10 + 5 = 15$ is returned as the output.

Example 2:

input1 : 2

input2 : 1

Output : 2

Explanation:

Bruce cannot make any new paint combinations since the existing paints are (1) and (2) only, and therefore **2** is returned as the output.

```
1 // Read only region st
2
3
4
5 int numberOfPaints(int n)
6 {
7     // Read only region
8     // Write code here
9
10 }
```

